DATA 606 Data Project Proposal

AUTHOR
Patricio Romero

PUBLISHED
October 21, 2026

Data Preparation

The first step is to load the required packages and import the NYCHA electricity dataset.

▶ Code

The dataset is imported into R and stored in an object named `nycha`.

Research question

How is electricity consumption associated with electricity cost among NYCHA housing developments in New York City, and how does this relationship vary across boroughs?

This question is limited to the NYCHA billing records included in the dataset. The results cannot be generalized to all buildings in New York City.

Cases

The cases are monthly electricity billing records for NYCHA housing developments. Each row represents an electricity record for a development during a specific billing period.

▶ Code

Summary of NYCHA Cases

Total Cases	Total Developments	Total Boroughs
553666	381	7

The output shows the total number of billing records, developments, and boroughs represented in the dataset.

Data collection

The data were collected from administrative electricity billing records for New York City Housing Authority developments. The records contain electricity consumption and cost information associated with developments, boroughs, and billing periods.

Type of study

This is an observational study. The researcher did not assign electricity usage, electricity cost, borough, or billing period. Therefore, the analysis can describe associations but cannot establish causation.

Data Source

The data were obtained from NYC Open Data.

Dataset: Electric Consumption and Cost
Publisher: New York City Housing Authority
Source: NYC Open Data — Electric Consumption and Cost

The original file is stored in:

```
Data/Electric_Consumption_And_Cost_(2010_-_Sep_2025)_20261018.csv
```

Describe your variables

The following chunk selects and prepares the variables used in the analysis.

▶ Code

```
Rows: 553,666
Columns: 6
$ development_name <chr> "SHEEPSHEAD BAY", "SHEEPSHEAD BAY", "SHEEPSHEAD BAY",…
$ borough          <chr> "BROOKLYN", "BROOKLYN", "BROOKLYN", "BROOKLYN", "BROO…
$ revenue_month    <chr> "2025-09", "2025-09", "2025-09", "2025-09", "2025-09"…
$ consumption_kwh  <dbl> 33001, 36068, 30969, 40235, 33956, 26999, 41264, 3057…
$ current_charges  <dbl> 5106.45, 5581.02, 4791.96, 6225.81, 5254.21, 4177.71,…
$ kwh_charges      <dbl> 1523.76, 1665.38, 1429.94, 1857.78, 1567.86, 1246.63,…
```

`development_name` and `borough` are qualitative variables. `revenue_month` is a time variable. `consumption_kwh`, `current_charges`, and `kwh_charges` are quantitative variables.

The main outcome variable is `current_charges`, and the main comparison variable is `consumption_kwh`. Borough will be used to compare the records across locations. The scatter plot includes a visual linear trend line to help describe the direction of the association. The trend line is used for visualization and is not presented as an inferential regression model. If a regression model were used, `current_charges` would be the dependent variable.

Relevant summary statistics

The following chunk calculates summary statistics for electricity consumption and electricity cost.

▶ Code

Summary Statistics for Electricity Consumption

Observations	Mean Consumption (MWh)	Median Consumption (MWh)	Standard Deviation (MWh)	Minimum Consumption (MWh)	Maximum Consumption (MWh)
553,666	30.22	11.44	48.26	0	2,788.77

▶ Code

Summary Statistics for Electricity Cost

Observations	Mean Cost (USD)	Median Cost (USD)	Standard Deviation (USD)	Minimum Cost (USD)	Maximum Cost (USD)
553,666	$4,339.03	$2,259.86	$6,302.43	-$25,622.21	$424,298.64

The first table reports the number of observations, mean, median, standard deviation, minimum, and maximum for electricity consumption. The second table reports the same statistics for electricity cost. Separating the variables makes the results easier to read and compare.

Summary statistics by borough

▶ Code

Electricity Consumption and Cost by Borough

Borough	Observations	Average Consumption (MWh)	Median Consumption (MWh)	Average Cost (USD)	Median Cost (USD)
STATEN ISLAND	4,143	107.75	63.20	$14,132.55	$12,689.84
BRONX	125,657	34.23	12.60	$4,917.68	$2,540.16
MANHATTAN	141,415	33.70	7.60	$4,884.52	$2,441.42
BROOKLYN	212,402	25.56	11.76	$3,613.67	$2,100.87
QUEENS	64,276	23.70	14.96	$3,539.93	$1,867.22

This table compares the typical electricity consumption and electricity cost among the boroughs.

Frequency of qualitative variables

The following table reports the number of records for each borough. This summarizes the qualitative variable used in the analysis.

▶ Code

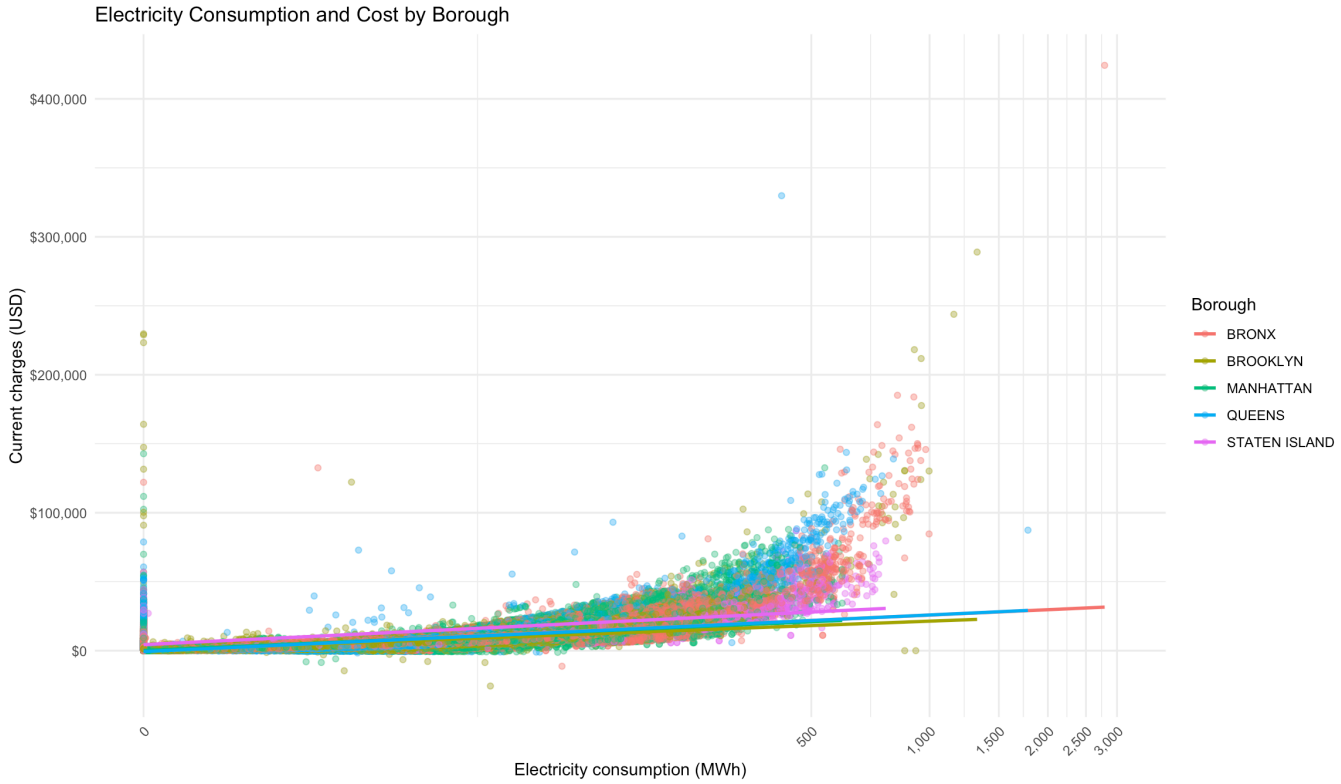
Number of Records by Borough

Borough	Observations
BROOKLYN	212402
MANHATTAN	141415
BRONX	125657
QUEENS	64276
STATEN ISLAND	4143
NON DEVELOPMENT FACILITY	3729
FHA	2044

This table shows how many electricity records are available for each borough.

Visualization: Electricity consumption and cost

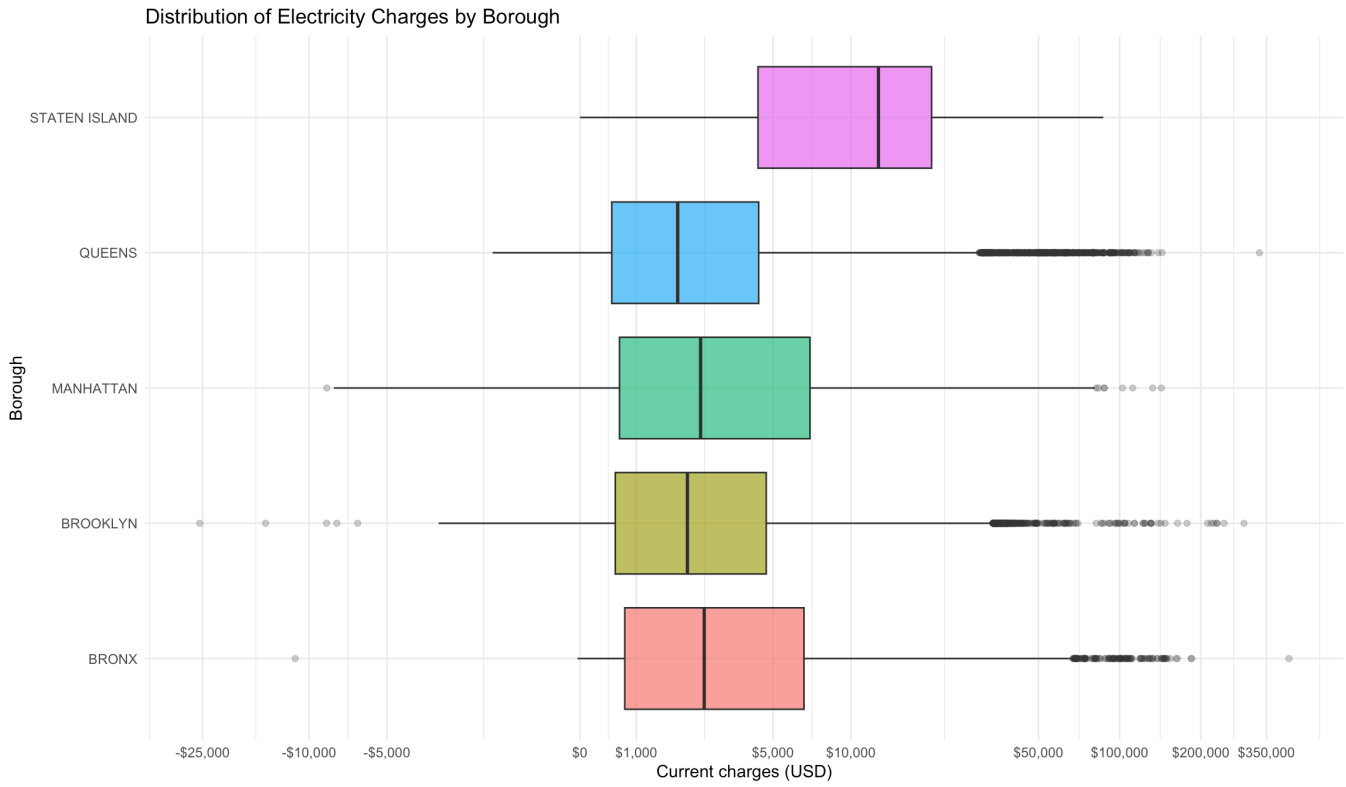
▶ Code



Each point represents one electricity billing record. The plot examines whether higher electricity consumption tends to be associated with higher electricity cost.

Visualization: Electricity cost by borough

▶ Code



This boxplot compares the median, variability, and possible outliers of electricity charges across boroughs.